Q1) Connect an Excel workbook to a SQL Server database using Power Query and load the 'Restaurants' table (assume it contains columns like name, cuisine, rating) into a new worksheet.

Click Get Data $\rightarrow$ From Database $\rightarrow$ From SQL Server Database.

Specify Server and Database Details:In the pop-up window, enter the Server Name and Database Name.

Select Import as the Data Connectivity mode and click OK.Authenticate:Select the authentication method

(e.g., Windows or Database username/password) and click Connect.

Select the Table:In the Navigator window, locate and check the box next to the Restaurants

table (displaying columns such as name, cuisine, and rating).Load to Worksheet:

Click Load (or click Transform Data $\rightarrow$ Close & Load).

Power Query will import the Restaurants table into a new worksheet formatted as an Excel Table.

Q2) Set up a parameterized query in Power Query that allows you to filter the 'Restaurants' data by cuisine type (e.g., only show 'Chinese' or 'South Indian') based on a value entered in an Excel cell.

Hint: Use Power Query's 'Manage Parameters' feature and reference the cell value in your SQL query.

Step 1: Create a Named Range for the Input Cell in ExcelIn your Excel worksheet,

type a default cuisine type (e.g., Chinese) into a specific cell (for example, cell B1).

Select that cell (B1), go to the Name Box (located to the left of the formula bar), type a meaningful name like CuisineInput,

and press Enter.

Step 2: Create a Parameter in Power QueryOpen the Power Query Editor (go to Data > Get Data > Launch Power Query Editor

or open an existing query).

On the Home tab, click Manage Parameters $\rightarrow$ New Parameter.

Configure the parameter:Name: CuisineParamType: TextCurrent Value: Chinese (matching your Excel cell)Click OK.

Step 3: Import and Link the Excel Cell to the ParameterIn Excel, select your named cell (CuisineInput)

and go to Data $\rightarrow$ From Table/Range to load it into Power Query.

In the Power Query Editor, right-click the single data value in the preview table and

select Drill Down (this converts the table into a text scalar value).

Rename this query to SelectedCuisine.

Go back to your parameter settings or formula to ensure CuisineParam evaluates to

or takes its value from SelectedCuisine (or use SelectedCuisine directly as your filter variable).

Step 4: Modify the SQL Query to Use the ParameterSelect your main Restaurants query connected to SQL Server.

Open the Advanced Editor (or look at the formula bar for the Source step).

Update the SQL query string in the Sql.Database function to include a WHERE clause that references your parameter:

Code snippet= Sql.Database("YourServerName", "YourDatabaseName", [Query="SELECT name, cuisine,

rating FROM Restaurants WHERE cuisine = '" & CuisineParam & "'"])

Click Done, then click Close & Load to return the filtered data to your worksheet.

Whenever you change the cuisine name in the Excel cell (CuisineInput) and click Refresh,

Power Query will dynamically pass that value to the SQL Server query and return only the matching records.

Q3) Enable dynamic data refresh in your Excel sheet so that when new records are added to the 'Restaurants' table in SQL Server, you can update the Excel data with a single click.

Open Connection Properties:

Go to the Data tab on the Excel ribbon.

Click Queries & Connections to open the side panel displaying your queries.

Right-click your Restaurants query (or the connection) and select Properties.

Configure Refresh Options:

In the Connection Properties dialog box, navigate to the Usage tab.

Check the box for Enable background refresh (this allows Excel to fetch data without locking up your sheet).

(Optional) If you want the data to update automatically upon opening the file, check Refresh data when opening the file.

Click OK to save the changes.

Perform a Single-Click Refresh:

Whenever new records are added to the Restaurants table in SQL Server, you can update your Excel sheet instantly with a single click.

Simply go to the Data tab and click Refresh All (or right-click anywhere inside your Excel table and click Refresh).

Power Query will re-run the query against SQL Server and pull in the latest records.

Q4) Modify your Power Query connection to include only restaurants with a rating above 4.0, and display the filtered results in a dashboard-style Excel sheet with a table and a bar chart.

Open the Power Query Editor:

Go to the Data tab and click Queries & Connections.

Right-click your Restaurants query and select Edit.

Apply the Rating Filter:

Option A (via SQL): Open the Advanced Editor and update your SQL WHERE clause to include the rating condition:

= Sql.Database("YourServerName", "YourDatabaseName", [Query="SELECT name, cuisine, rating

FROM Restaurants WHERE cuisine = '" & CuisineParam & "' AND rating > 4.0"])

Option B (via Power Query UI): Click the drop-down arrow on the Rating column header, select Number Filters > Greater Than..., enter 4.0, and click OK.

Click Close & Load to update the worksheet.

Part 2: Create a Dashboard-Style Excel Sheet with a Bar Chart

Format the Table:

Ensure your imported data is formatted as an official Excel Table (click inside the data, go to Insert > Table if not already formatted).

Apply a clean visual style using the Table Design tab (e.g., matching header colors, alternating row shading).

Insert a Bar Chart:

Select the data columns you want to visualize (e.g., select the name and rating columns, or summarize by cuisine).

Go to the Insert tab on the ribbon.

Click Recommended Charts or choose Column/Bar Chart (e.g., a clustered column or horizontal bar chart).

Design the Dashboard Layout:

Move the chart next to or above your data table within the worksheet.=

Go to the Chart Design tab to pick a professional color palette and layout.

Hide gridlines if desired (View tab > uncheck Gridlines) to give it a clean, dashboard-like appearance.

Q5) Use ChatGPT or Copilot to help you write a parameterized SQL query for Power Query that fetches all restaurants within a certain rating range (e.g., between 3.5 and 5.0), then implement and test it in your Excel workbook.

```
SELECT name, cuisine, rating

FROM Restaurants

WHERE rating BETWEEN @MinRating AND @MaxRating
```

Create Min and Max Parameters in Power Query

Go to Data > Queries & Connections > Right-click your query and select Edit (opens Power Query Editor).

Go to the Home tab > Manage Parameters > New Parameter:

Parameter 1: Name it MinRating, Type: Decimal Number, Current Value: 3.5.

Parameter 2: Name it MaxRating, Type: Decimal Number, Current Value: 5.0.

Click OK.

: Update the Advanced Editor with the Range Query

Select your Restaurants query in the left panel.

Go to the Home tab and click Advanced Editor.

Modify the Sql.Database function to concatenate both parameters into your SQL WHERE clause:

= Sql.Database("YourServerName", "YourDatabaseName", [Query="SELECT name, cuisine, rating

FROM Restaurants WHERE rating BETWEEN " & Number.ToText(MinRating) & " and " & Number.ToText(MaxRating)])

Click Done.

Click Close & Load to return to your Excel worksheet.

Test the Setup:

Go back into Power Query (Data > Queries & Connections > Edit), change the MinRating parameter value (e.g., change from 3.5 to 4.5), and click Close & Load.

Alternatively, link these parameters to specific Excel cells using named ranges (similar to the cuisine example earlier) so users can change the range directly on the sheet.

Click Refresh to verify that only restaurants falling within the specified rating range are loaded into your workbook.